Orange Digital Center

INSTANT

# Marketing Analytics

DATA ANALYSIS HACKATHON

Dataset file name: " *MarketingAnalyticsProject.bak"*

Contains 5 tables: Customers, Products, Customers_Journey, Customers_Reviews, Engagement_Data

## Scenario & Business Emails

- Email 1 – Marketing Manager (Jane Doe)

**Subject: Request for Data Analysis to Improve Marketing Strategy**

*Hi Data Analyst,*

I'm Jane Doe, the Marketing Manager at ShopEasy. Despite new marketing campaigns, we've seen reduced engagement and low conversion rates. Our marketing spend has increased but ROI is below expectations. We need your analysis to understand what's going wrong and how to improve.

*Best regards,*
*Jane Doe*
*Marketing Manager – ShopEasy*

- Email 2 – Customer Experience Manager (John Smith)

**Subject: Request for Analysis of Customer Feedback**

*Hi Data Analyst,*

I'm John Smith, the Customer Experience Manager at ShopEasy. We've noticed a drop in customer satisfaction and engagement. We've collected a large amount of customer reviews and social media comments. Please analyze this data to understand customer pain points and help us improve engagement and conversion rates.

*Thank you,*
*John Smith*
*Customer Experience Manager – ShopEasy*

## Project Overview

ShopEasy, an online retail business, is facing reduced engagement, lower conversion rates, and higher marketing expenses with unclear customer sentiment. Your task is to act as a Data Analyst Team and deliver a full analysis identifying root causes, cleaning data, building KPIs, and presenting insights with business recommendations.

## Project Goals

1. Increase Conversion Rates – Identify drop-off points and suggest improvements.
2. Enhance Customer Engagement – Analyze what content drives higher engagement.
3. Improve Customer Feedback Scores – Detect common themes and suggest improvements.

## Key Performance Indicators (KPIs)

- Conversion Rate – % of website visitors who make a purchase.
- Customer Engagement Rate – Interaction with content (clicks, likes, comments).
- Average Order Value (AOV) – Average amount spent per transaction by each customer.
- Customer Feedback Score – Average rating from reviews.
- ROI / Cost per Acquisition (CPA) – Marketing cost vs. conversions.

## Project Requirements

1. SQL

- Restore and explore the .bak database.
- Detect and clean data issues (duplicates, nulls, formatting, …).

2. Power BI

- Build an interactive Power BI dashboard showing KPIs and insights.
- Include filters (by product, campaign, or time).
- Use visuals like conversion funnel, engagement trend, sentiment overview.
- Publish live dashboard and embed into PowerPoint for presentation.

3. Python (Optional)

- Perform EDA and trend analysis using pandas & matplotlib/seaborn.
- Generate additional insights or predictive models (bonus).
- Optionally automate PowerPoint reporting.

4. Final Presentation

- Present insights, visuals, and recommendations clearly.
- Explain data cleaning (SQL) and analysis (Power BI or Python).
- Include business problem, steps, KPIs, findings, and recommendations.

## Tips for Success

- Document every SQL step clearly.
- Keep visuals simple and business-oriented.
- Connect insights to KPIs and recommendations.
- Ensure PowerPoint has live visuals from Power BI.
- Work as a team and assign clear roles.

## Criteria for Evaluation :

Data Accuracy & Preparation (25%)

- Correct handling of nulls, duplicates, and formatting issues
- Valid and consistent data after cleaning (SQL-based)
- Dataset ready for KPI calculation and analysis

Analysis & Insights (30%)

- Clear identification of conversion drop-off points
- Analysis of customer engagement and campaign performance
- Root cause analysis linked directly to KPIs
- Actionable business insights (not observations only)

Visualization Quality (25%)

- Clear and accurate KPI visuals (Conversion, AOV, CPA, Engagement)
- Proper use of funnels, trends, and comparisons
- Clean and professional Power BI design

Dashboard Usability (15%)

- Effective use of filters (product, campaign, time)
- Easy navigation and logical dashboard structure
- Quick access to key insights and KPIs

Presentation (5%)

- Clear summary of findings and recommendations
- Professional and structured delivery
- Live Power BI visuals embedded in PowerPoint

Bonus Points (Max +5%)

- Predictive Analysis (Python): Simple, justified trend or prediction model
- Advanced Analysis: Sentiment analysis or cohort-based insights

Notes

- Bonus points do not compensate for incorrect KPIs or weak analysis.
- Incorrect KPI calculation results in score reduction.